

3 + 4

Trigonometry 2 Paper 2

TRIG GRAPHS

QUESTION 1

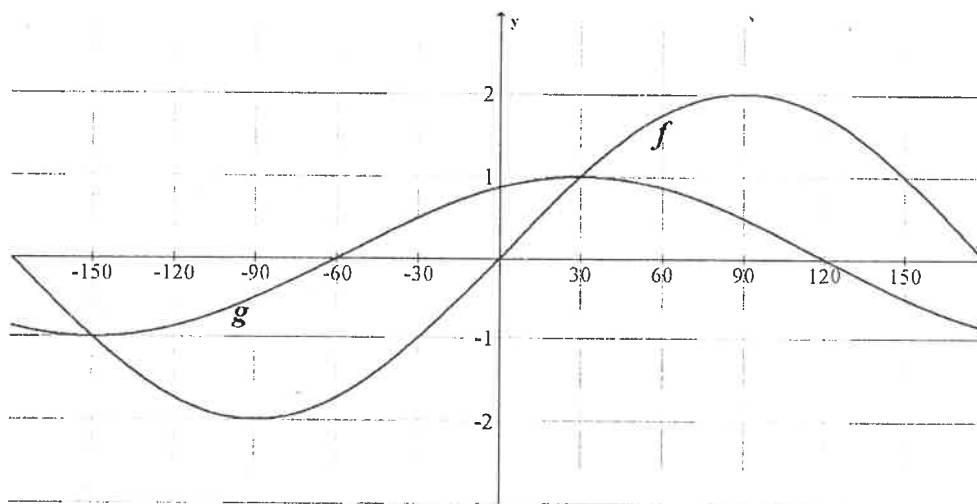
Given $f(x) = 2\cos x$ and $g(x) = \sin(x + 30^\circ)$

- 1.1. Write down the amplitude of f . (1)
- 1.2. Sketch the graph of f and g on the same set of axes $x \in [-180^\circ; 90^\circ]$. (5)
- 1.3. Determine the value(s) of x for which $f(x) - g(x) \geq 0$ for $x \in [-180^\circ; 90^\circ]$. (2)
- 1.4. Give TWO possible ways of transforming the graph of f such that its y-intercept is at the origin. (2)

[10]

QUESTION 2

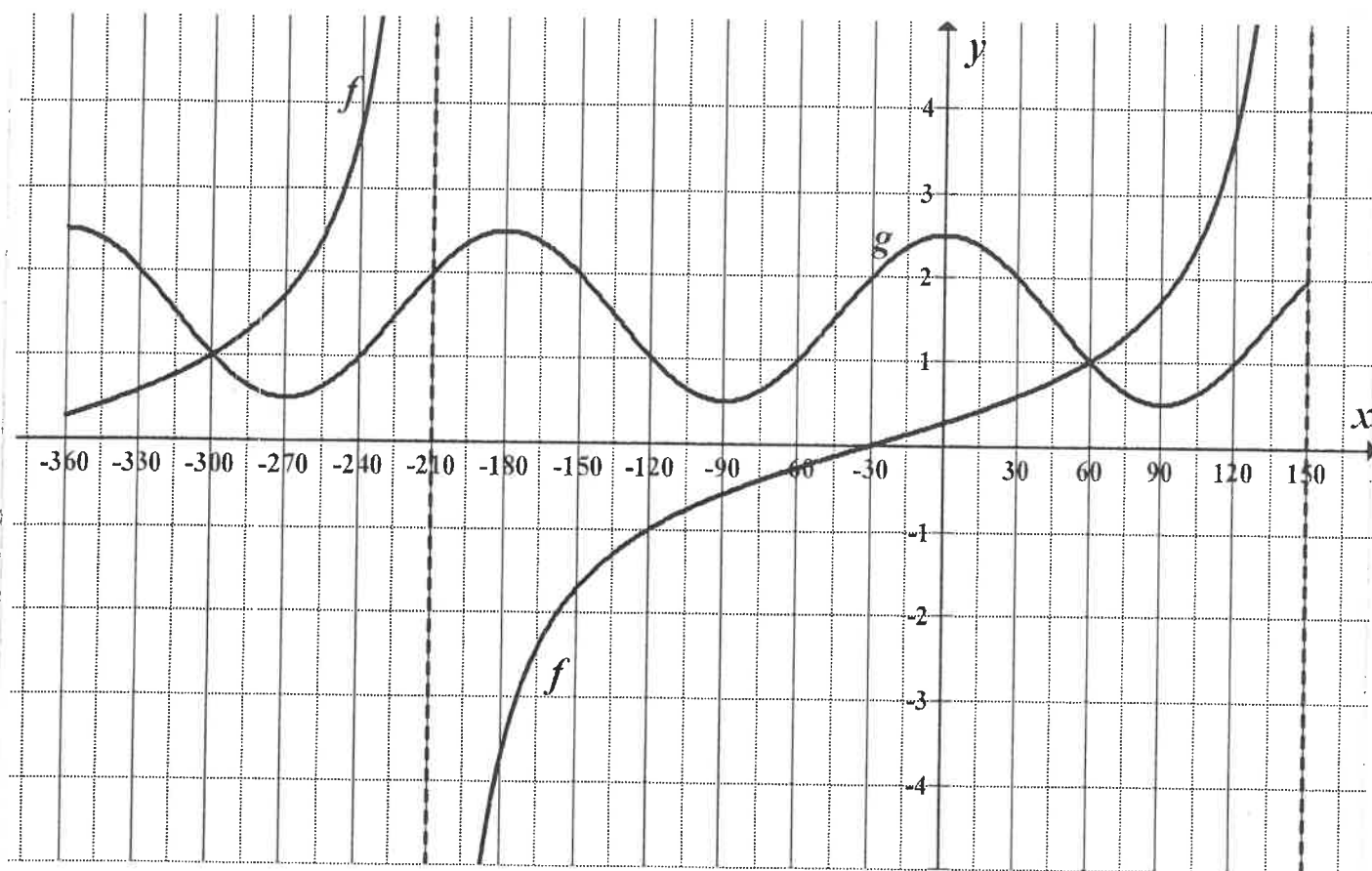
The graphs of $f(x) = 2\sin x$ and $g(x) = \cos(x - 30^\circ)$, for $x \in [-180^\circ; 180^\circ]$ are drawn below.



- 2.1. Write down the range of $g(x) + 2$. (2)
- 2.2. Write down the period of $f(\frac{1}{2}x)$. (2)
- 2.3. Use your graphs to write down the values of x for which $\frac{f(x)}{g(x)} = 1$, for $x \in [-180^\circ; 180^\circ]$. (2)
- 2.4. For which values of x , $x \in [-180^\circ; 180^\circ]$, is $f'(x) \cdot g(x) \leq 0$. (3)
- 2.5. Use the graphs to determine the values of x in the interval $[-180^\circ; 180^\circ]$ for which $\sqrt{3} \cos x + \sin x = 4 \sin x$.

QUESTION 3

The graphs of the functions $f(x) = \tan\left(\frac{x}{2} + 15\right)$ and $g(x) = \cos(2x) + \frac{3}{2}$ on the interval $-360^\circ \leq x \leq 150^\circ$ are shown in the diagram below.



3.1. By using the graphs drawn above:

3.1.1. Write down the x -values of the points of intersection for $f(x) = g(x)$ for $-360^\circ \leq x \leq 150^\circ$ (2)

3.1.2. Determine, the period of $f(x)$. (1)

3.1.3. Determine, the range of $g(x)$. (2)

3.2. Determine the value(s) of x for which:

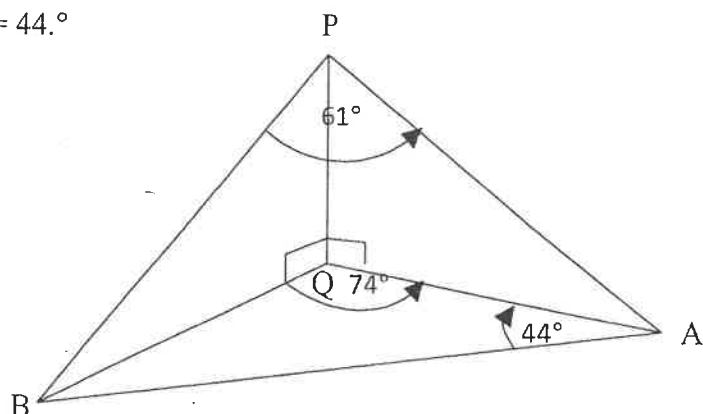
3.2.1. $g(x) - f(x) = 2$ for $x \in [-210^\circ; 0^\circ]$ (2)

3.2.2. $f(x) \cdot g(x) \geq 0$ for $x \in [-360^\circ; 150^\circ]$ (2)

SIN - COS - AND AREA RULES * Remember the proofs of these rules.

QUESTION 4

A vertical pole PQ is supported by two chains PB and PA which are 32 metres and 25 metres respectively in length. A, B and Q are in the same horizontal plane. $\widehat{BPA} = 61^\circ$, $\widehat{BQA} = 74^\circ$ and $\widehat{QAB} = 44^\circ$.



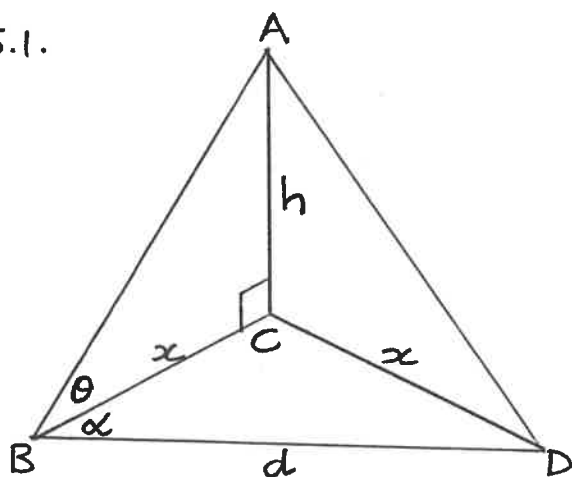
Calculate the lengths of:

4.1. BQ (5)

4.2. AQ (2)

QUESTION 5

5.1.

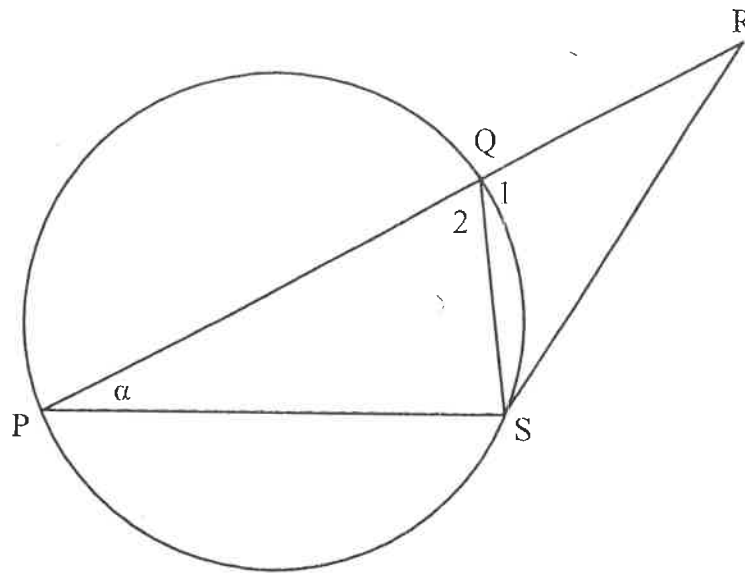


In the sketch, BCD is a triangle in a horizontal plane.

AC is a vertical line.

Prove that $h = \frac{d \tan \theta}{2 \cos \alpha}$.

- 5.2. In the figure, the diameter PQ of the circle is produced to R. S is a point on the circumference such that $QR = QS = x$. P and S are joined and $\angle QPS = \alpha$.



5.2.1. Prove that : $SR = x\sqrt{2(1 + \sin \alpha)}$ (4)

5.2.2. If $SR = 5\sqrt{3}$ and $x = 5$, show that $PQ = 10$ (4)

QUESTION 6

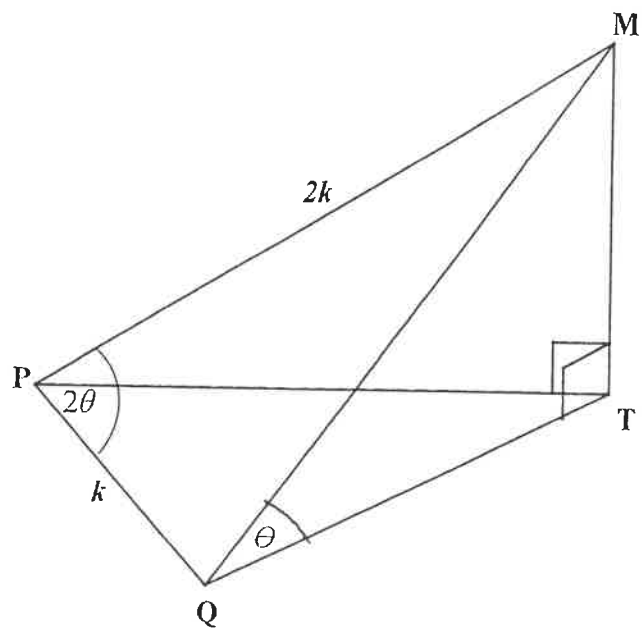
In the diagram alongside, MT is a vertical structure.

P, Q and T are three points in the same horizontal plane.

The angle of elevation of M from Q is θ .

$PQ = k$ metres.

$PM = 2 PQ$, $\angle MPQ = 2\theta$



6.1. Prove that :

$$MQ = k\sqrt{1 + 8\sin^2 \theta} \quad (5)$$

6.2. Find the value of MQ, rounded off to the nearest metre if $k = 139,5$ m and $\theta = 42^\circ$. (2)

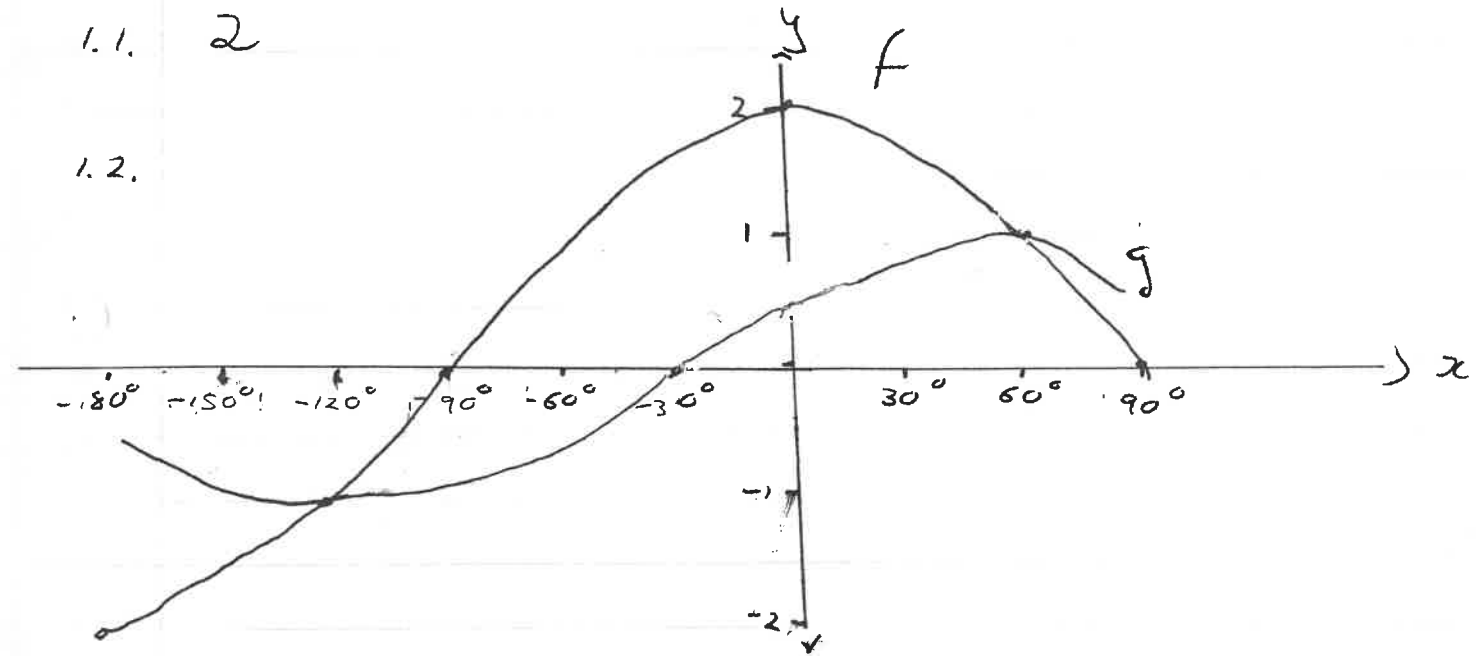
Trigonometry 2: Solutions

Trig Graphs

Question 1

1.1. 2

1.2.



1.3. $-120^\circ \leq x \leq 60^\circ$

1.4. $2 \cos(x - 90^\circ)$ shift 90° right
 $2 \cos(x + 90^\circ)$ shift 90° left

Question 2:

2.1. $y \in [1; 3]$

2.2. 720°

2.3. $x = 30^\circ$ OR $x = -150^\circ$

2.4. $-90^\circ \leq x \leq -60^\circ$ OR $90^\circ \leq x \leq 120^\circ$

2.5 $\sqrt{3} \cos x + \sin x = 4 \sin x$

$$\frac{\sqrt{3}}{2} \cos x + \frac{1}{2} \sin x = 2 \sin x$$

$$\cos 30^\circ \cos x + \sin 30^\circ \sin x = 2 \sin x$$

$$\cos x \cos 30^\circ + \sin x \sin 30^\circ = 2 \sin x$$

$$\cos(x - 30^\circ) = 2 \sin x$$

$\therefore x = 30^\circ$ OR $x = -150^\circ$

Question 3:

3.1.1. $x = -300^\circ$ OR $x = 60^\circ$

3.1.2. 360°

3.1.3. $y \in \left[\frac{1}{2}; \frac{5}{2}\right]$

3.2.1. $x = -120^\circ$ OR $x = -30^\circ$

3.2.2. $-360^\circ \leq x < -210^\circ$ OR $-30^\circ \leq x < 150^\circ$

Sin- Cos- and Area Rules

4.1. In ΔPBA :

$$\begin{aligned} AB^2 &= PB^2 + AP^2 - 2(PB)(AP)\cos P \\ &= (32)^2 + (25)^2 - 2(32)(25)\cos 61^\circ \end{aligned}$$

$$AB = \underline{29.55 \text{ m}}$$

In ΔABQ :

$$\begin{aligned} \frac{BQ}{\sin A} &= \frac{AB}{\sin Q} \\ \frac{BQ}{\sin 44^\circ} &= \frac{29.55}{\sin 74^\circ} \\ BQ &= \frac{29.55 \sin 44^\circ}{\sin 74^\circ} \\ &= \underline{21.35 \text{ m}}, \end{aligned}$$

4.2. In ΔABQ

$$\begin{aligned} \hat{B} &= 180^\circ - 74^\circ - 44^\circ \quad (\text{int } \angle \text{'s of } \Delta) \\ &= \underline{62^\circ} \end{aligned}$$

$$\begin{aligned} \frac{AQ}{\sin B} &= \frac{AB}{\sin Q} \\ AQ &= \frac{AB \sin B}{\sin Q} \\ &= \frac{29.55 \sin 62^\circ}{\sin 74^\circ} \\ &= \underline{27.14 \text{ m}}, \end{aligned}$$

Question 55.1. In ΔBCD :

$$\begin{aligned}\hat{B} &= \hat{D} = \alpha & \angle\text{'s opp equal sides} \\ \hat{C} &= 180^\circ - 2\alpha & \text{int } \angle\text{'s of } \Delta\end{aligned}$$

$$\begin{aligned}\frac{BC}{\sin D} &= \frac{BD}{\sin C} \\ \frac{x}{\sin \alpha} &= \frac{d}{\sin(180^\circ - 2\alpha)} \\ x &= \frac{d \sin \alpha}{\sin 2\alpha} \\ &= \frac{d \sin \alpha}{2 \sin \alpha \cdot \cos \alpha} \\ &= \frac{d}{2 \cos \alpha}\end{aligned}$$

In ΔABC :

$$\begin{aligned}\tan \theta &= \frac{h}{x} \\ h &= x \tan \theta \\ &= \frac{d \tan \theta}{2 \cos \alpha}\end{aligned}$$

$$5.2.1 \quad \angle PSQ = 90^\circ$$

$\angle$ in a semi circle

$$\hat{Q}_1 = 90^\circ + \alpha$$

ext $\angle$ of Δ

In Δ SQR:

$$\begin{aligned} SR^2 &= QS^2 + QR^2 - 2QS \cdot QR \cos \hat{Q}_1 \\ &= x^2 + x^2 - 2x \cdot x \cos (90^\circ + \alpha) \\ &= 2x^2 - 2x^2 (-\sin \alpha) \\ &= 2x^2 + 2x^2 \sin \alpha \\ &= 2x^2 (1 + \sin \alpha) \end{aligned}$$

$$\begin{aligned} SR &= \sqrt{2x^2 (1 + \sin \alpha)} \\ &= x \sqrt{2(1 + \sin \alpha)} \end{aligned}$$

$$5.2.2 \quad 5\sqrt{3} = 5 \sqrt{2(1 + \sin \alpha)}$$

$$\sqrt{3} = \sqrt{2(1 + \sin \alpha)}$$

$$3 = 2(1 + \sin \alpha)$$

$$\frac{3}{2} = 1 + \sin \alpha$$

$$\frac{1}{2} = \sin \alpha$$

$$\therefore \alpha = 30^\circ$$

In Δ PQS:

$$\sin \alpha = \frac{QS}{PQ}$$

$$\sin 30^\circ = \frac{5}{PQ}$$

$$\frac{1}{2} = \frac{5}{PQ}$$

$$\underline{PQ = 10}$$

Question 6:6.1. In ΔMPQ :

$$\begin{aligned}
 MQ^2 &= MP^2 + PQ^2 - 2MP \cdot PQ \cdot \cos \hat{P} \\
 &= (2k)^2 + (k)^2 - 2(2k)(k) \cos 2\theta \\
 &= 4k^2 + k^2 - 4k^2 \cos 2\theta \\
 &= k^2 (5 - 4 \cos 2\theta) \\
 &= k^2 (5 - 4(1 - 2\sin^2 \theta)) \\
 &= k^2 (5 - 4 + 8\sin^2 \theta) \\
 &= k^2 (1 + 8\sin^2 \theta)
 \end{aligned}$$

$$\begin{aligned}
 MQ &= \sqrt{k^2 (1 + 8\sin^2 \theta)} \\
 &= k \sqrt{1 + 8\sin^2 \theta}
 \end{aligned}$$

$$\begin{aligned}
 6.2. \quad MQ &= k \sqrt{1 + 8\sin^2 \theta} \\
 &= 139.5 \sqrt{1 + 8\sin^2 (42^\circ)} \\
 &= \underline{298.60 \text{ m}}
 \end{aligned}$$